

Uncertainty Quantification of Spherical Measurements in 3D LiDAR

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Introduction

LiDAR systems are fundamental to modern perception, autonomous navigation, and environmental surveillance [1]. However, they face a significant physical challenge: the extreme speed of light leads to quantization errors in pulsed Time-of-Flight (ToF) measurements [2]. In maritime robotics, these effects can obscure small floating objects such as plastic debris, reducing detection reliability [3].

In this work, each LiDAR measurement is represented in spherical coordinates as

$$\mathbf{p}_{i,j,t} = [\rho_{i,j,t}, \theta_{i,j,t}, \phi_{i,j,t}]^T, \quad (1)$$

where $\rho_{i,j,t}$ denotes the radial distance, $\phi_{i,j,t}$ the elevation angle, and $\theta_{i,j,t}$ the azimuth angle at horizontal index i , vertical channel j , and time t . Instead of modeling measurements as discrete points, the proposed approach represents them as bounded spatial regions derived from temporal variations across multiple frames, improving robustness to noise and quantization effects.

Spherical Inclusion Box

To characterize measurement uncertainty, a temporal sequence of n frames acquired in a static environment is analyzed. For each pixel

(i, j) , the temporal mean radial distance $\bar{\rho}_{i,j}$ and standard deviation $\sigma_{i,j}$ are computed. A conservative global uncertainty bound is then defined as

$$\Sigma_{\max} = \sup_{i,j}(\sigma_{i,j}), \quad (2)$$

where $\Sigma_{\max}$ represents the maximum observed radial deviation across the sensor. The radial uncertainty interval is defined as

$$[\rho]_{i,j} = [\bar{\rho}_{i,j} - \kappa \Sigma_{\max}, \bar{\rho}_{i,j} + \kappa \Sigma_{\max}], \quad (3)$$

where κ is a coverage factor, typically set to $\kappa = 3$ for a 99.7% confidence interval.

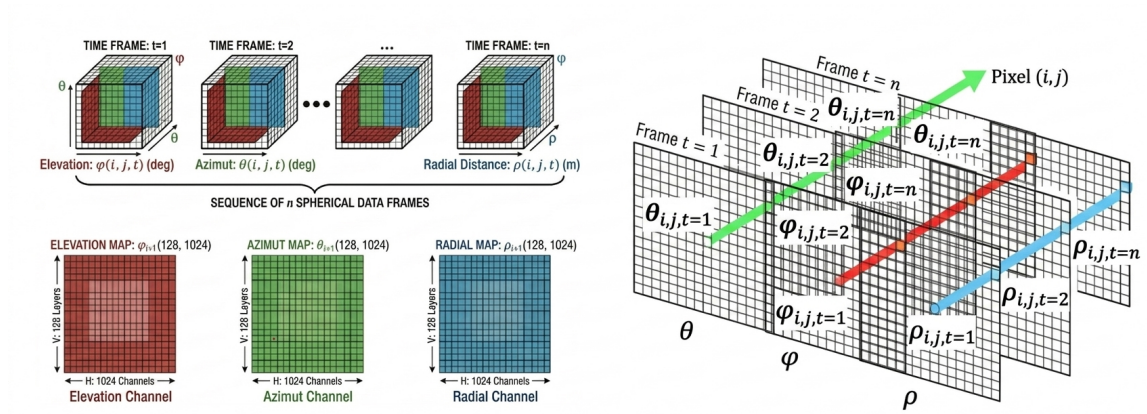


Figure 1: [Left] Spherical LiDAR data structure and temporal acquisition sequence for $t = 1, \dots, n$. [Right] Spatiotemporal aggregation highlighting the pixel-wise extraction of measurement traces.

Each measurement is then represented as a *Spherical Inclusion Box*

$$[\mathbf{p}]_{i,j} = [\rho]_{i,j} \times [\theta]_{i,j} \times [\phi]_{i,j} \quad (4)$$

where the angular intervals are bounded using the sensor quantization limits δ_θ and δ_ϕ extracted from the sensor metadata. This formulation provides a compact geometric representation of LiDAR uncertainty while remaining independent of the sensor operating environment.

Experimental Results

Experiments were conducted in a static environment using an *Ouster OS1-128* sensor (128×1024 points per frame). To evaluate measurement stability, individual detector channels were analyzed over a 20-minute recording at a sampling frequency of 10 Hz, resulting in a dataset of 12 000 frames¹. This extensive temporal sequence allows for a robust characterization of the sensor’s noise profile.

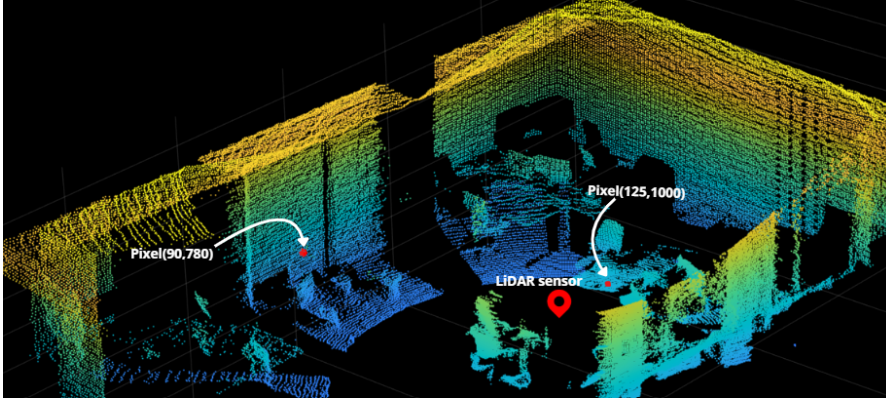


Figure 2: Point cloud of a static cluttered indoor environment. Red markers denote detector channels selected for temporal range analysis.

As illustrated in Fig. 2, we characterize the stochastic variation of radial measurements by observing static targets over time. The resulting temporal distributions were modeled using Gaussian fitting (see Fig. 3). As can be noticed, pixel $\mathbf{p}_{90,780,t}$ exhibits a standard deviation of $\sigma = 0.0075$ m, whereas $\mathbf{p}_{125,1000,t}$ reaches $\sigma = 0.0133$ m. This non-uniformity justifies the adoption of a conservative global bound, $\Sigma_{\max}$, to ensure that the defined uncertainty intervals remain reliable across the entire point cloud.

By analyzing the full point cloud over multiple frames, we established a global radial supremum of $\Sigma_{\max} \approx 0.0133$ m. Applying a coverage factor of $\kappa = 3$ (99.7% confidence), each measurement is transformed into a *Spherical Inclusion Box* $[\mathbf{p}]_{i,j}$ which replaces dis-

¹Dataset available at <https://drive.univ-littoral.fr/s/6BPeBDfPx92WQjC>

crete points with guaranteed spatial occupancy regions:

$$[\mathbf{p}]_{i,j} = [\bar{\rho}_{i,j} \pm 0.04] \times [\theta_{i,j} \pm \delta_\theta] \times [\phi_{i,j} \pm \delta_\phi] \quad (5)$$

where the angular uncertainty corresponds to the sensor's half-resolution limits: $\delta_\theta = \frac{360^\circ}{2 \times 1024} \approx 0.175^\circ$ and $\delta_\phi = \frac{45^\circ}{2 \times 128} \approx 0.175^\circ$.

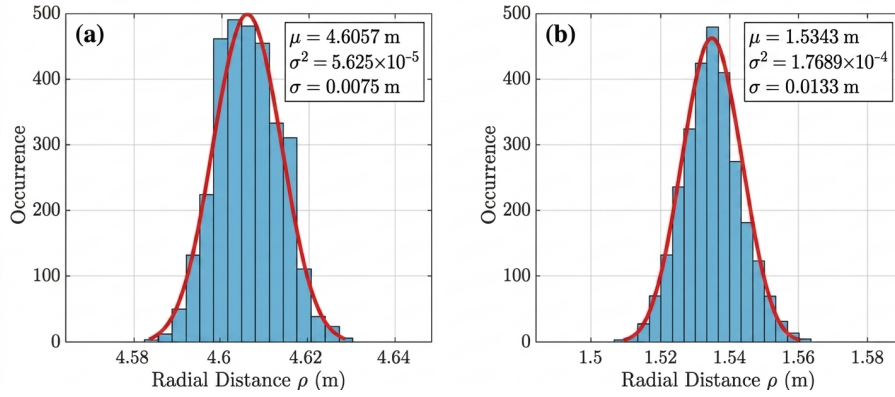


Figure 3: Radial distance uncertainty characterization via Gaussian fitting for selected points $\mathbf{p}_{90,780}$ and $\mathbf{p}_{125,1000}$.

References

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